

DEEP-DREAM: A DOCUMENT-FIRST TEMPORAL CONCEPT GRAPH FOR AGENTIC LONG-TERM MEMORY

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ABSTRACT

Long-term memory for an agent is an information-allocation problem: a system must find the relevant past under a bounded read budget while preserving the evidence needed to verify a claim. We study a document-first temporal concept graph that separates these roles. Deep-Dream stores immutable source spans and organizes their observations, concepts, relations, and versions into a graph used for low-cost addressing; an agent then expands the address and progressively reads source evidence until its uncertainty is reduced. This design treats concept navigation as a compact code and source text as the retained evidence channel, rather than forcing one write-time summary to serve every future query. Progressive alignment is a posterior-refinement process: uncertain mentions may remain separate, while later content, relational neighborhoods, and update order can support non-destructive merges. On a 767-question MemoryAgentBench sample, extending the available evidence path from direct retrieval to a memory-tool agent to source-grounded verification yields Overall scores of 0.324, 0.407, and 0.651; the same ordering appears in an independent engine generation (0.303, 0.376, 0.669). The pattern is an answer-path systems comparison, not a single-factor causal estimate. LongMemEval, LoCoMo, and MEME expose complementary properties: temporal and multi-session access, multi-hop consolidation, and the current absence of deletion semantics. The Align+ revision reduces observed ingest calls by 46–65% and same-name duplicate families, while retaining comparable sampled quality rather than improving every score. Offline channel replay bounds the derived graph’s static retrieval effect to ± 2.4 points, consistent with a role as non-authoritative navigation. These results support an agentic memory principle: compress the address, preserve the source, and defer evidence allocation to the query.

1 INTRODUCTION

An agent that remembers for a long time faces an information-allocation problem. It must locate a relevant past event among an expanding history, but it must also preserve the wording, qualifier, and update that make a future answer correct. A fixed summary is efficient to read, yet it commits a reconstruction decision before the future question is known. Keeping every document in every prompt preserves information, yet makes retrieval and attention scale with the entire history. The central question is therefore not how to store more text, but how to separate low-cost addressing from evidence that remains verifiable.

We define Deep-Dream as a document-first temporal concept graph for agentic long-term memory. Documents and exact source spans are authoritative. Episodes, entity and relation concepts, observations, and version links form a derived state that answers which source should be opened next. The graph is therefore an address layer, not a replacement for the source. A query-time agent can search, traverse, read, and verify; an evidence gate accepts only source identifiers surfaced by the current read path. An alignment error can consequently reduce coverage, but it cannot make an unobserved derived sentence count as read evidence.

The representation follows three principles. First, the graph has one state axis: processing/observation time. Each new input appends an observation and may create a new version; event dates remain part of the observed content rather than a second state timeline. Second, concepts are stable handles while observations retain the exact text that produced them. Third, an uncertain first alignment is allowed to remain an over-split family. New observations and relational neighborhoods can later reduce identity uncertainty and redirect pointers without deleting evidence. This makes revision safer than an early merge that contaminates one address with two entities.

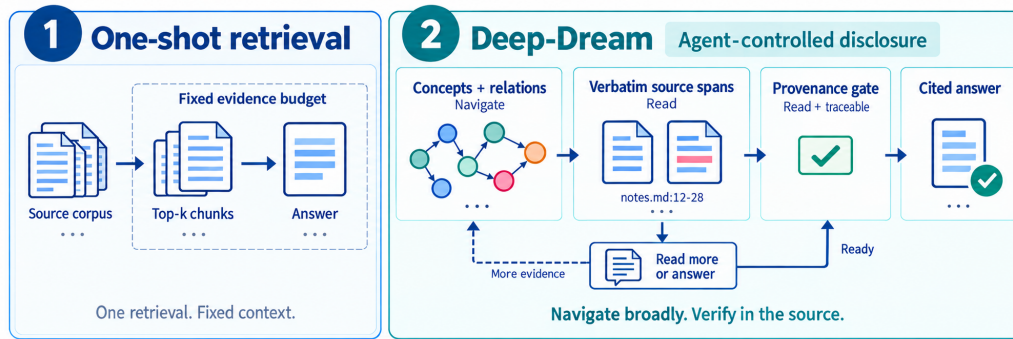


Figure 1: The memory interface as an information-allocation problem. A fixed retrieval payload commits the answer context before the question is resolved. Deep-Dream uses a temporal concept graph to address source documents, then lets the agent expand and verify verbatim spans under a query-dependent read budget.

These principles admit an information-theoretic account. For a query q , let A be the address produced by concept and relation navigation and E_C the source spans opened from candidate set C . The answer-relevant information decomposes into address information and residual evidence information. A summary-only system fixes a rate-distortion point at write time; Deep-Dream defers the evidence rate to the agent’s read policy. The agent can stop at a compact concept neighborhood for an easy query, or spend additional budget reading episodes, neighboring relations, and source files when the posterior uncertainty remains high.

We turn this account into testable predictions. The answer path should matter most for questions requiring several mentions or an update qualifier. Temporal and multi-session benchmarks should reveal whether old and new observations remain jointly addressable. Multi-hop benchmarks should reveal whether relation paths connect evidence across episodes. Deletion benchmarks should expose the boundary between preserving history and suppressing history. Finally, if alignment is progressive rather than a one-shot merge, a revised engine should reduce duplicate families and construction cost even when its aggregate accuracy is not uniformly higher.

We evaluate these predictions on four complementary instruments. MemoryAgentBench (MAB) compares direct retrieval, a memory-tool loop, and source-grounded verification on the same sampled questions and frozen libraries. LongMemEval measures long-range updates and multi-session access under paired full-ingest protocols. LoCoMo measures binary-judged single-hop, multi-hop, open-domain, and temporal recall. MEME provides a negative boundary for deletion, cascade, and absence. Published memory-system results are used as protocol-labeled context, with answerer, judge, subset, and context-length differences preserved.

Our contributions are:

- **A temporal concept-graph memory model.** Documents are the authority; concepts, relations, observations, and versions provide a rebuildable address state ordered by observation time.
- **An information-theoretic explanation.** We formalize progressive alignment, the two-rate address/evidence decomposition, and query-time rate-distortion selection, with explicit assumptions and non-claims.
- **An agentic evidence protocol.** The agent controls search, graph expansion, source reading, verification, and stopping while a deterministic gate enforces provenance reachability.
- **A capability-axis evaluation.** MAB, LongMemEval, LoCoMo, and MEME test answer-path access, temporal updates, relational consolidation, and deletion boundaries rather than collapsing them into one score.
- **A calibrated comparison.** We explain gains and failures against published memory systems without cross-protocol subtraction, and report the Align+ cost/consolidation trade-off and negative results.

The claims are deliberately bounded. The answer-path ladder is a systems comparison rather than a single-factor intervention; the MAB result is a 767-question sample; the temporal model is a representation and revisability principle rather than a completed current/as-of/withdrawal benchmark; and no same-model full-context anchor is available because the Kimi-K3 endpoint was retired.

2 RELATED WORK

Memory benchmarks. LoCoMo and LongMemEval test long conversations, temporal updates, and multi-session reasoning (Maharana et al., 2024; Wu et al., 2025). MEME adds deletion, cascade, and absence checks (Jung et al., 2026). MemoryAgentBench (MAB) evaluates retrieval, test-time learning, long-range understanding, and selective forgetting in one official scoring framework (Hu et al., 2025); we use it as the primary instrument and keep its task-level metrics visible.

Structured memory. MemGPT, A-MEM, Mem0, HippoRAG-2, Cognee, MIRIX, and Zep/Graphiti improve persistence through summaries, notes, or graphs (Packer et al., 2023; Xu et al., 2025; Chhikara et al., 2025; Gutiérrez et al., 2025; Markovic et al., 2025; Wang & Chen, 2025; Rasmussen et al., 2025). Deep-Dream shares the need for compact navigation but separates the authority boundary: a derived node points to a source span; it is not itself answer evidence.

Information-theoretic and associative memory. The information bottleneck formalizes a compact representation that preserves information relevant to a target variable (Tishby et al., 2000), while rate-distortion theory characterizes the cost of lossy reconstruction (Shannon, 1959). Associative-memory models explain content-addressable retrieval from partial cues (Ramsauer et al., 2021), and probabilistic record linkage separates false-link and missed-link errors (Fellegi & Sunter, 1969). We use these ideas as an analysis of the memory interface: the overlay is a low-rate address code, exact spans are the retained evidence channel, and delayed entity merges trade a tolerable missed-link cost against catastrophic false-link contamination.

Retrieval, long context, and provenance. Retrieval-augmented generation selects evidence (Lewis et al., 2020), while long-context models can lose information when relevant spans are far from the attention window’s ends (Liu et al., 2024). Attribution work commonly checks citations after generation (Gao et al., 2023); our gate checks reachability at runtime. It does not prove entailment, and we report that limitation explicitly.

3 DEEP-DREAM

3.1 MEMORY MODEL

Let D be a source document and e_i an episode extracted from one of its spans. Deep-Dream keeps D , versions, episodes, and exact source offsets as authoritative data. An entity family f is a stable search handle; each observation $o_{f,i}$ stores what a source asserted, when it was asserted, and where it can be reread. Relations use the same family/assertion split. Processing documents are hidden until publication, and active observations are separated from historical ones. Derived updates therefore add or merge pointers without overwriting source text.

3.2 INGESTION AND ALIGNMENT

The Remember pipeline performs heading-aware chunking, validated entity/relation extraction, conservative family alignment, window-batch adjudication, and scope-end convergence. In Align+, entity alignment runs concurrently across windows. Candidate equivalence groups are queued during ingestion; family merges are deferred until the scope has no concurrent writers, preserving references used by relation construction. Early over-splitting is recoverable because observations retain their source links. The combined Align+ revision has fewer same-name duplicate families (340→124 on MAB; 482→99 on LME) and 46–65% fewer observed ingest calls per document (Section 4.5); these measurements do not isolate convergence from the accompanying changes.

3.3 ANSWER PATHS AND EVIDENCE CONTRACT

All three paths use the same question stream, frozen library, answerer, scorer, and scope eligibility. They differ in the accessible answer path:

1. **Direct retrieval:** lexical, semantic, and graph channels are fused into top- k source spans; a separate answerer sees the payload once.
2. **Memory-tool agent:** a bounded loop searches the derived records and provenance tools, then submits the surfaced evidence to a separate answerer.
3. **Source-grounded agent:** the overlay materializes a source scope; the agent reads verbatim files with ordinary tools, verifies the answer, and submits the spans it actually read.

These are different systems rather than a single-factor intervention. The answer-path matrix in the appendix makes the differences explicit; the channel replay in Section 4.8 is the single-factor diagnostic.

The evidence gate runs without an LLM. It accepts an identifier only if it was returned by a tool call for the current question, belongs to an eligible active scope, and retains a source-level pointer. This is provenance reachability, not semantic entailment: a model can still cite a reachable but irrelevant span. The runtime records tool observations, evidence IDs, latency, exit status, and code/runtime hashes for every item.

Candidate records combine lexical, semantic, concept, relation, and graph-neighborhood channels by reciprocal-rank fusion, resolve to episodes and spans, and add neighboring turns when needed; in the source-grounded path the overlay only chooses the scope, so an addressing error causes missing coverage rather than a fabricated citation.

3.4 PROGRESSIVE DISCLOSURE IN INFORMATION-THEORETIC TERMS

Let $X_{1:t}$ be the observations extracted from the first t source spans and let Z denote their latent entity assignments. The persistent state is $M_t = (F_t, O_t, G_t)$: concept families F_t , an append-only source-linked observation store O_t , and a relation graph G_t . Ingestion adds evidence and may redirect a family, but does not delete an observation:

$$\begin{aligned} O_{t+1} &= O_t \cup \{(X_{t+1}, \text{span}_{t+1})\}, \\ I(Z; O_{1:t+1}) &= I(Z; O_{1:t}) + I(Z; X_{t+1} \mid O_{1:t}) \geq I(Z; O_{1:t}). \end{aligned} \quad (1)$$

Thus the first alignment is a provisional partition, not a lossy commitment. When a merge is uncertain, creating a new family preserves the observations needed to revisit it. Under the idealized i.i.d. feature model in Appendix A.13, the probability that a true merge remains undetected decreases exponentially with pooled evidence; in the implementation, relation-endpoint co-occurrence supplies additional discriminative features and later becomes a retrieval edge.

A two-rate memory code. For a query q , let A be the compact address returned by concept families and relation paths, and let E_C be the exact source spans opened from the selected candidate set C . The information available to the answerer decomposes as

$$I(Y; A, E_C \mid Q = q) = I(Y; A \mid Q = q) + I(Y; E_C \mid A, Q = q), \quad (2)$$

where Y is the answer-relevant latent fact. The first term is a low-rate navigation code; the second is the residual evidence exposed only when the agent follows an address. This is the architecture’s central separation: concept and relation records compress the search space, while the append-only source store retains the bits that a future question may need. A summary-only memory replaces S by $\tilde{S} = f(S)$ and, by the data-processing inequality, satisfies $I(Y; \tilde{S} \mid Q) \leq I(Y; S \mid Q)$. Deep-Dream instead keeps S addressable and uses derived records as an information bottleneck for navigation, in the sense of Tishby et al. (2000); it does not commit to a fixed answer distortion at write time. Resolving a pointer to its immutable span has source distortion zero, while the address code can remain small enough to avoid the long-context dilution documented by Liu et al. (2024).

This also clarifies why the graph is useful even when a static recall ablation is nearly null. The graph need not contain the answer; it only has to increase the conditional information of the next read, $I(Y; E_{C_{k+1}} \mid A, C_{\leq k}, Q)$. Its value is therefore measured at the interface between addressing and source reading, rather than by treating a graph node as a replacement for the source. This view is consistent with content-addressable memory models (Ramsauer et al., 2021) and graph-indexed long-term memory systems (Gutiérrez et al., 2025).

Progressive alignment as posterior refinement. Let $R_{1:t}$ denote relation-endpoint and temporal features observed alongside $X_{1:t}$. Under a calibrated likelihood model, the posterior uncertainty of an entity assignment obeys

$$\begin{aligned} H_{t+1} &= H(Z \mid X_{1:t+1}, R_{1:t+1}) \\ &= H_t - I(Z; X_{t+1}, R_{t+1} \mid X_{1:t}, R_{1:t}) \leq H_t. \end{aligned} \quad (3)$$

The inequality formalizes the reason that a first mention need not be aligned perfectly: later mentions can only add conditional information in the idealized model. The relation term can be helpful even when the name is ambiguous, because $I(Z; R_{t+1} \mid X_{1:t+1}, R_{1:t})$ captures evidence supplied by co-occurring endpoints and update order. A conservative linker can therefore treat an uncertain match as a new family and postpone the irreversible operation. This is the same two-error tradeoff

formalized for probabilistic record linkage by Fellegi & Sunter (1969); Deep-Dream assigns the larger cost to a false merge because it contaminates every future address that points to that family.

At read time, the overlay is an address map rather than a reconstruction. A query first selects a concept neighborhood and relation paths, then progressively discloses source spans:

$$C_0(q) \subseteq C_1(q) \subseteq \dots \subseteq S, \quad a_{k+1} = \rho(q, M_t, C_{\leq k}, u_k), \quad (4)$$

where S is the immutable source store, ρ is the agent’s read policy, and u_k is its current uncertainty. The overlay reduces the search space at the macro level; each selected family resolves to real episodes and text at the micro level. Because every cited span belongs to S , reconstruction distortion is zero by construction; the only failure mode is addressing loss, namely that the relevant span never enters C_k . The agent therefore chooses its own rate–distortion point: it can stop after a compact concept lookup or spend more tokens reading and checking source files, while the gate preserves provenance at every step.

For a token budget B , the ideal query policy can be written as

$$\rho_B^*(q) = \arg \max_{C: \text{cost}(C) \leq B} I(Y; E_C \mid A, Q = q). \quad (5)$$

The deployed agent approximates this policy by reading until its uncertainty is low enough or the evidence gate cannot be satisfied. If R_q denotes that the relevant span is addressed and V_q denotes correct verification after that span is read, then

$$\Pr[\hat{Y} = Y \mid q] \geq \Pr[R_q \mid q] \Pr[V_q \mid R_q, q]. \quad (6)$$

The overlay primarily raises the first factor; source access and progressive disclosure raise the second. The deterministic gate additionally makes provenance violations impossible under its contract, although it does not guarantee semantic entailment. This decomposition predicts the observed pattern: the largest gains occur on multi-mention and fact-consolidation tasks, where both address recall and evidence sufficiency are needed, while a graph-only offline replay can remain nearly unchanged.

4 EXPERIMENTS

We test five memory capabilities instead of reducing long-term memory to one score. Table 4 maps each capability to a benchmark and a prediction; all protocol labels and negative results are retained.

4.1 PROTOCOL AND SCOPE

Primary benchmark. MAB contributes 767 sampled questions from 10 scopes and 59 documents (the full set has 3,671 questions). The sample covers eight tasks and uses the official scorer at commit 455306d. Kimi-K3 answers all paths and both engine generations: *Base* is the initial engine and *Align+* the revised engine. We report raw Overall and, when needed for cross-system context, an adjusted estimate (59–61).

Other protocols. LongMemEval full-ingest uses paired $n = 25/25/27$ samples for the three paths (the source-grounded track has two extra questions). The separate 500-question LME-S result is 84.60% with Qwen3.7-plus cross-judging. LoCoMo-1540 follows Mem0’s binary- J protocol with Kimi-K3 and is diagnostic, not a shared leaderboard. MEME tests deletion and absence. Full sampling and judge details are in Section A.3.

4.2 RQ1: DOES ADAPTIVE EVIDENCE ACQUISITION IMPROVE ANSWERS?

Table 1 reports the six frozen MAB runs. Source-grounded retrieval is highest, especially on MCC and FC-MH.

On these questions, Align+ scores 0.324/0.407/0.651 for Direct retrieval, Memory-tool agent, and Source-grounded agent; Base scores 0.303/0.376/0.669. The paired bootstrap intervals are +0.083 [+0.015, +0.149] and +0.244 [+0.214, +0.305]. These paths also change tools, prompts, answerer arrangement, and compute, so the comparison is not a one-factor causal test. Source grounding raises MCC from 2 to 46 and FC-MH from 3 to 70. The singleton $n=1$ ∞ Bench-Sum row changes from 21.2 to 0.0; excluding it gives 75.5 for Align+ and 74.7 for Base. The two engines therefore show a cost/quality trade-off, not a ranking.

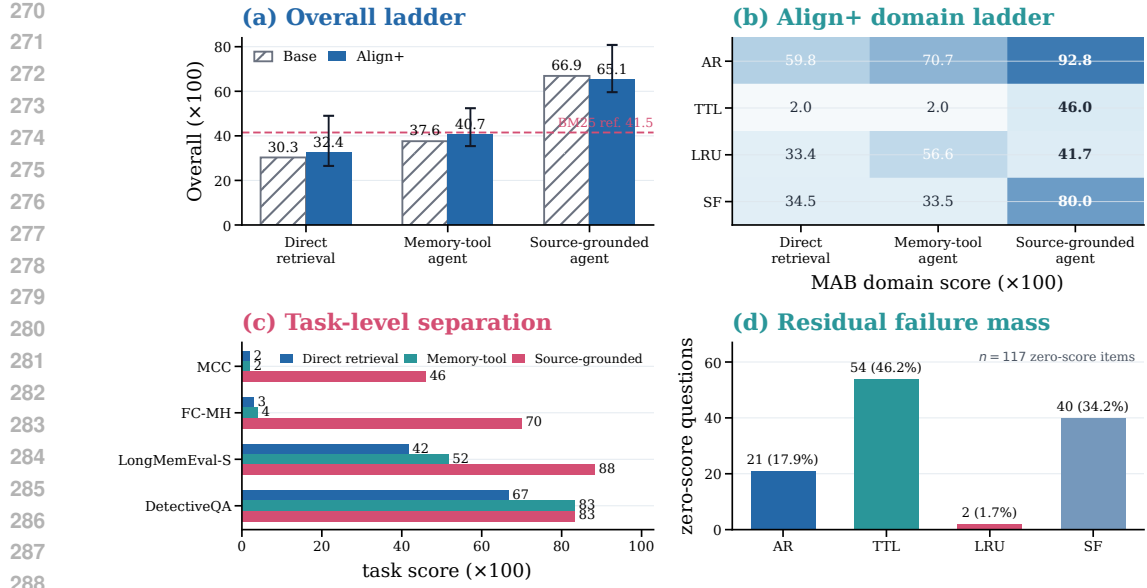


Figure 2: Capability-axis view of the MemoryAgentBench answer-path result (official scorer, $n=767$, Kimi-K3 answerer). (a) Overall scores for Deep-Dream Base (hatched) and Align+ (solid), with paired-bootstrap 95% CIs on Align+ and a published BM25 reference band. (b) Align+ domain scores show where progressive source access matters: associative retrieval (AR), temporal/test-time learning (TTL), long-range understanding (LRU), and fact consolidation (SF). (c) Representative task rows expose the largest path separations, especially MCC and FC-MH. (d) The residual zero-score taxonomy reports all 117 failed items by domain. The published BM25 value uses a different answerer, judge, and subset; no subtraction is implied.

Task / domain	Direct retrieval		Memory-tool agent		Source-grounded agent		Align+ gain
	Base	Align+	Base	Align+	Base	Align+	Align+ gain
AR: Associative Retrieval							
SH-QA	54.0	48.0	84.0	72.0	94.0	<u>92.0</u>	44.0
LongMemEval-S*	26.7	41.7	43.3	51.7	90.0	<u>88.3</u>	46.6
EventQA	<u>90.3</u>	89.7	<u>90.7</u>	88.3	98.0	98.0	8.3
TTL: Temporal and Test-time Learning							
MCC, icl_clinic150	4.0	2.0	3.0	2.0	<u>43.0</u>	46.0	44.0
LRU: Long-range Understanding							
∞ Bench-Sum [†]	0.0	0.0	<u>28.4</u>	29.8	21.2	0.0	0.0
DetectiveQA	50.0	<u>66.7</u>	50.0	83.3	83.3	83.3	16.6
SF: Fact Consolidation							
FC-SH	<u>68.0</u>	66.0	67.0	63.0	90.0	90.0	24.0
FC-MH	2.0	3.0	4.0	4.0	<u>67.0</u>	70.0	67.0
Overall ($n = 767$)	30.3	32.4	37.6	40.7	66.9	<u>65.1</u>	32.7

Table 1: MemoryAgentBench on the shared 767-question sample; scores are official component scores $\times 100$. Bold/underlined entries are best/second-best within each task, the last column is the within-protocol Align+ path gap, * marks sampled LongMemEval-S, and [†] marks the singleton ∞ Bench-Sum row.

4.3 RQ2: DOES A TEMPORAL, VERSIONED STATE HELP UPDATES AND SESSIONS?

LongMemEval asks whether an earlier observation and a later correction remain available. The paired tracks use the same question IDs and answerer, but $n = 25/25/27$ is directional. The independent 500-question LME-S score is 84.60% with Qwen3.7-plus cross-judging; it is a separate protocol, not a common leaderboard.

The largest contextual margin is on temporal and knowledge-update questions, where old and new source spans must remain addressable. Session-preference questions remain a boundary because they

require latent user-state modeling. We use this benchmark to test temporal access, not to claim a universal temporal advantage.

4.4 RQ3: DO CONCEPT AND RELATION PATHS SUPPORT MULTI-HOP CONSOLIDATION?

On the Mem0-compatible LoCoMo-1540 protocol, Deep-Dream reaches 95.19% weighted binary- J with Kimi-K3; the same answers score 94.22% with a Qwen3.7-plus cross-judge. Relative to Mem0’s published row, the gaps are +42.47 multi-hop, +39.82 temporal, +30.25 single-hop, and +7.28 open-domain. Models and protocols differ, so these numbers diagnose task structure rather than establish a cross-protocol ranking.

The pattern is consistent with the design: multi-hop questions need concept aggregation, relation traversal, and a return to exact source spans; temporal questions also need update order and old versions. Open-domain questions often need only one episode. The full protocol and legacy-fingerprint caveat are in the appendix.

4.5 RQ4: DOES PROGRESSIVE ALIGNMENT MAKE CONSTRUCTION REVISABLE AND ECONOMICAL?

Base: commit identity as evidence arrives. Base resolves each window sequentially and consolidates it immediately; scope-end convergence is disabled. This is simple, but an early miss leaves duplicate families and an early false merge can contaminate one address with another entity’s observations.

Align+: separate evidence capture from identity commitment. Align+ ingests windows concurrently, keeps candidate groups, and merges compatible families only after a scope closes. Uncertain evidence can therefore remain split and later converge, while source observations stay immutable. In plain terms: record first and decide with more context later.

In paired runs, calls fall from 609 to 329 per MAB document (−46%) and from 63.8 to 22.0 per LME document (−65.5%). Duplicate families fall from 340 to 124 (MAB) and 482 to 99 (LME); MCC and FC-MH each rise by 0.03 with intervals +0.030 [−0.070, +0.130] and +0.030 [−0.070, +0.130]. Concurrency, convergence, retries, timeouts, and throughput fixes are bundled, so this is evidence of a cost/revisability trade-off, not a component ablation. Details are in Appendix A.2.

4.6 RQ5: WHERE DOES THE APPEND-ONLY DESIGN FAIL?

MEME marks the current limit: deletion accuracy is 0% on 100 episodes, while cascade and absence real accuracy are 46.34 and 59.23. The store keeps prior observations and links but has no query-time suppression or deletion propagation. TTL is also below the strongest full-context MAB rows. Temporal revision and deletion are separate capabilities.

4.7 CROSS-SYSTEM POSITIONING BY AUTHORITY AND PROTOCOL

Table 2 gives protocol-labeled context for MAB, LoCoMo, and LongMemEval-S.

Deep-Dream uses the graph for addressing and source spans for authority. The cleanest length-matched, judge-free MAB comparisons are SH-QA (92 vs. 87) and DetectiveQA (83.3 vs. 77.5). EventQA and FactConsolidation use different context lengths (our FC scopes are 6K; published rows average 262K), and Overall 59–61 vs. 49.6 uses different models and subsets. These numbers are context, not a unified SOTA claim.

4.8 MECHANISM DIAGNOSTICS AND OPERATING COST

On 210 development/gate questions, changing only derived channels gives Recall-any@10 of 72.38, 72.38, 71.43, and 72.38%. The bound is ± 2.4 points and Recall@1, MRR, and nDCG agree (recall@1 25.7/24.8/25.7/26.2, MRR@10 40.2/40.1/40.5/41.0, nDCG@10 42.0/42.2/42.2/42.7). This offline null does not remove the online overlay, which still materializes source scopes, selects versions, and expands links. The evidence gate rejects an unsurfaced identifier; deployed rates are 11/767 questions in the deployed Align+ Memory-tool agent run had an evidence submission rejected as unsurfaced (9/767 in Base); 9 of the 11 resubmitted tool-surfaced evidence inside the loop and the remaining 2 ended at the step cap and were answered from tool-surfaced context; zero rejections on the LME runs. The full arm table is in the appendix.

(a) MemoryAgentBench (official scorer)						
System / answerer	AR	TTL	LRU	SF	Overall	
Full-context / GPT-4o	58.1	50.0	54.9	32.5	48.8	
Full-context / GPT-4o-mini	49.2	48.6	46.2	25.0	42.2	
Full-context / GPT-4.1-mini	71.8	46.2	49.1	20.5	46.9	
Full-context / Gemini-2.0-Flash	65.1	46.4	41.6	16.5	42.4	
Full-context / Claude-3.7-Sonnet	59.7	53.9	62.2	22.5	49.6	
BM25 RAG / GPT-4o-mini	60.5	44.5	35.6	25.5	41.5	
HippoRAG-2 / GPT-4o-mini	65.1	35.8	36.2	29.5	41.6	
MIRIX / GPT-4.1-mini	63.0	35.7	40.5	11.5	37.7	
MemGPT / GPT-4o-mini	34.3	40.8	22.4	15.5	28.3	
Zep / GPT-4o-mini	37.5	37.5	16.2	5.0	24.0	
Mem0 / GPT-4o-mini	32.6	21.2	20.7	10.0	21.1	
Deep-Dream / Direct retrieval (Kimi-K3)	59.8	2.0	33.4	34.5	32.4	
Deep-Dream / Memory-tool agent (Kimi-K3)	70.7	2.0	56.6	33.5	40.7	
Deep-Dream / Source-grounded agent (Kimi-K3)	92.8	46.0	41.7	80.0	65.1	

(b) LoCoMo binary-J (%; Mem0 protocol)					
System / judge	Single-hop	Multi-hop	Open-domain	Temporal	Weighted
Mem0 / GPT-4o-mini	67.13	51.15	72.93	55.51	62.14
Mem0 ^g / GPT-4o-mini	65.71	47.19	75.71	58.13	61.36
Zep / GPT-4o-mini	61.70	41.35	76.60	49.31	≈56.32
LangMem / GPT-4o-mini	62.23	47.92	71.12	23.43	≈52.08
OpenAI / GPT-4o-mini	63.79	42.92	62.29	21.71	≈51.10
A-Mem* / GPT-4o-mini	39.79	18.85	54.05	49.91	≈38.95
Deep-Dream / Kimi-K3	97.38	93.62	80.21	95.33	95.19
Deep-Dream / Qwen3.7-plus cross-judge	96.91	92.91	77.08	93.46	94.22

(c) LongMemEval-S (%)							
System / answerer	Overall	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowl. upd.	Sess. user
Full-context / GPT-4o-mini	55.4	30.0	81.8	36.5	40.6	76.9	81.4
Zep / GPT-4o-mini	63.8	53.3	75.0	54.1	47.4	74.4	92.9
Full-context / GPT-4o	60.2	20.0	94.6	45.1	44.3	78.2	81.4
Zep / GPT-4o	71.2	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream / Qwen3.7-plus	84.6	53.3	100.0	84.2	74.4	91.0	98.6

Table 2: Protocol-labeled cross-system results for MemoryAgentBench, LoCoMo, and LongMemEval-S; published rows are from Hu et al. (Hu et al., 2025), Chhikara et al. (Chhikara et al., 2025), and Rasmussen et al. (Rasmussen et al., 2025). Answerer, judge, subset, and protocol remain attached to each block; values are contextual, not a shared leaderboard.

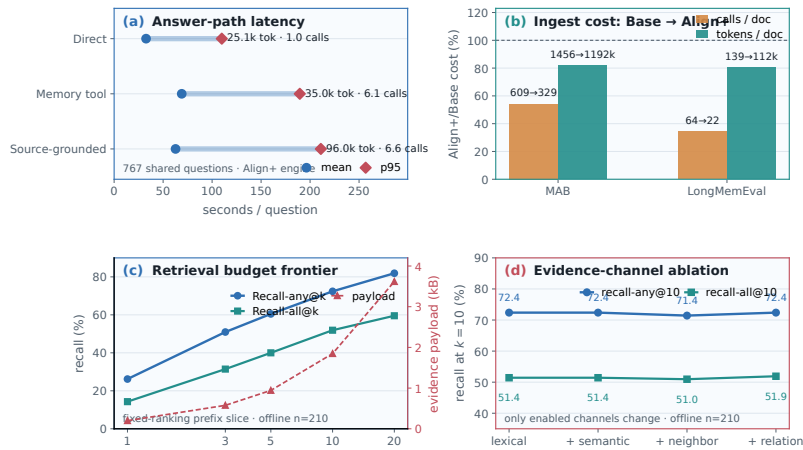


Figure 3: Operating cost and retrieval diagnostics: latency, Align+/Base ingest cost, retrieval budget, and source-channel ablation. Samples and bundled-version caveats are stated in the text.

4.9 NEGATIVE RESULTS RETAINED

Source-grounded still fails 34.9% of macro Overall; the residual taxonomy is 46.2% test-time learning, 34.2% selective forgetting (30/10 FC-MH/FC-SH questions), 17.9% long-context reading, 1.7% long-

range understanding. Direct retrieval and Memory-tool agent tie on the small paired LongMemEval tracks. We retain these failures, the MEME deletion result, the LRU singleton flip, and the X7 null to show what addressing, verification, and deletion still miss.

5 ANALYSIS AND LIMITATIONS

The results support a simple reading. More source reading helps most on multi-mention and fact-consolidation tasks: Align+ Overall is 0.324, 0.407, and 0.651 across the three paths, with the same ordering in Base. Tools, prompts, answerer arrangement, and compute change together, so this is a systems comparison, not a causal ablation. LongMemEval updates and LoCoMo multi-hop questions fit the temporal and relation design; preference and open-domain questions need less of it. Align+ cuts ingest calls by 46–65%, reduces duplicate families, and adds 3 points on MCC and FC-MH, but does not win every score. MEME deletion accuracy is 0%, which is the main boundary.

The diagnostics give the same picture. X7 bounds the static effect of derived channels to ± 2.4 points offline, while the online source-grounded path still uses the overlay to materialize scopes, choose versions, and expand links. The gate only guarantees provenance reachability; it can miss a source, but an accepted citation points to a source surfaced for the question. This explains why graph-only replay is small while multi-mention source reading helps.

The design fits long-term memory because the needed granularity is unknown when data is written. Concepts and relations say where to look; append-only observations preserve what was said; later evidence can merge provisional families without deleting source text; and the agent spends more read tokens only when needed. The formulas in Section 3.4 are decompositions and policies, not measured entropy gains. Their idealized claims assume a calibrated observation model and i.i.d. separable entity features.

The evaluation is limited. MAB is a 767-question sample with one singleton LRU item; LongMemEval full-ingest uses $n = 25/25/27$; public rows use different answerers, judges, subsets, and context lengths. We lack a same-model full-context anchor and complete runtime ablations. Deletion, current-as-of suppression, and a dense TTL benchmark remain future work.

6 CONCLUSION

Deep-Dream is a document-first temporal concept graph for agentic long-term memory. Concepts and relations provide a compact address, immutable observations preserve evidence, and the agent reads more only when needed. The experiments link source verification to multi-mention gains and show Align+ construction savings, while MEME exposes the unresolved deletion boundary. All comparisons remain tied to their stated answerer, judge, subset, and protocol.

AI ASSISTANCE

We used LLM-based assistance, including OpenAI Codex, to draft, restructure, and edit the manuscript; prepare a Chinese reading version; search for related literature and bibliographic information; and assist with analysis and plotting code. Generative image tools were used to create and revise explanatory framework figures.

AI assistance also contributed to the information-theoretic interpretation of the architecture, the formulation of mathematical claims and their supporting arguments, and the discussion of experimental methodology and results. This included the treatment of progressive entity alignment, address-evidence separation, and query-dependent reading budgets. These arguments are conditional on the assumptions stated in the paper; AI assistance does not constitute independent proof verification or empirical validation.

No synthetic evaluation dataset was created as part of this paper-preparation workflow. Reported benchmark scores are drawn from recorded experimental runs rather than generated performance estimates. The authors remain responsible for the final text, citations, mathematical claims, code, figures, and empirical conclusions, including verification of AI-assisted material.

REFERENCES

- Herman Chernoff. A measure of asymptotic efficiency for tests of a hypothesis based on the sum of observations. *The Annals of Mathematical Statistics*, 23(4):493–507, 1952. doi: 10.1214/aoms/1177729330. URL <https://doi.org/10.1214/aoms/1177729330>.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*, 2025. URL <https://arxiv.org/abs/2504.19413>.
- Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969. doi: 10.1080/01621459.1969.10501049. URL <https://doi.org/10.1080/01621459.1969.10501049>.
- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 6465–6488, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.398. URL <https://aclanthology.org/2023.emnlp-main.398/>.
- Bernal Jiménez Gutiérrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. From rag to memory: Non-parametric continual learning for large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 21497–21515. PMLR, 2025. URL <https://proceedings.mlr.press/v267/gutierrez25a.html>.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating memory in llm agents via incremental multi-turn interactions. *arXiv preprint arXiv:2507.05257*, 2025. URL <https://arxiv.org/abs/2507.05257>.
- Seokwon Jung, Alexander Rubinstein, Arnas Uselis, Sangdoo Yun, and Seong Joon Oh. Meme: Multi-entity & evolving memory evaluation. *arXiv preprint arXiv:2605.12477*, 2026. URL <https://arxiv.org/abs/2605.12477>.
- Patrick S. H. Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pp. 9459–9474, 2020. URL <https://arxiv.org/abs/2005.11401>.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Parandekar, Danqi Chen, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638. URL <https://arxiv.org/abs/2307.03172>.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*,

pp. 13851–13870, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.747. URL <https://aclanthology.org/2024.acl-long.747/>.

Vasilije Markovic, Lazar Obradovic, Laszlo Hajdu, and Jovan Pavlovic. Optimizing the interface between knowledge graphs and llms for complex reasoning. *arXiv preprint arXiv:2505.24478*, 2025. URL <https://arxiv.org/abs/2505.24478>.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*, 2023. URL <https://arxiv.org/abs/2310.08560>.

Hubert Ramsauer, Bernhard Schöfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021. URL <https://arxiv.org/abs/2008.02217>.

Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory. *arXiv preprint arXiv:2501.13956*, 2025. URL <https://arxiv.org/abs/2501.13956>.

Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. In *IRE National Convention Record*, volume 7, pp. 142–163, 1959. URL <https://ieeexplore.ieee.org/document/5311476>.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. Alfworld: Aligning text and embodied environments for interactive learning. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=0IOX0YcCdTn>. ICLR 2021.

Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. *arXiv preprint physics/0004057*, 2000. URL <https://arxiv.org/abs/physics/0004057>.

Yu Wang and Xi Chen. Mirix: Multi-agent memory system for llm-based agents. *arXiv preprint arXiv:2507.07957*, 2025. URL <https://arxiv.org/abs/2507.07957>.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. Longmemeval: Benchmarking chat assistants on long-term interactive memory. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/d813d324dbf0598bbdc9c8e79740ed01-Abstract-Conference.html. ICLR 2025.

Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-mem: Agentic memory for llm agents. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-0593. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/19909c36f51abc4856b4560aff3d36d6-Abstract-Conference.html.

Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, et al. Bigcodebench: Benchmarking code generation with diverse function calls and complex instructions. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/a6a90bcc2aa470c3871b2d39a67d26e8-Abstract-Conference.html. ICLR 2025.

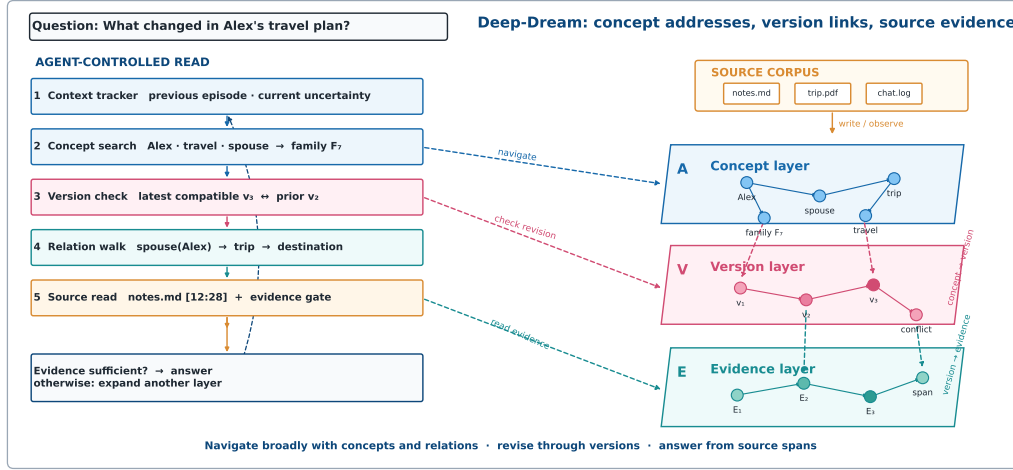


Figure 4: Deep-Dream separates agent-controlled progressive reading from a linked memory stack. The concept layer provides stable handles and relation neighborhoods; the version layer records observation-time revisions and conflicts; and the evidence layer resolves each address to an episode, chunk, and exact source span. During *write*, uncertain mentions may remain split and later converge; during *read*, the agent traverses concepts and relations, checks compatible versions, opens source evidence, and stops when the evidence gate is satisfied. Concept–version and version–evidence links make the representation revisable without changing the authority of the original span.

Property	Direct retrieval	Memory-tool agent	Source-grounded agent
Source access	top- <i>k</i> spans	derived records	verbatim sources
Tools	none	memory tools	read, grep, script
Answerer	separate	separate	verifying agent
Loop	none	bounded tool loop	bounded read/verify loop
Cost / question	25.1k tokens, 1.0 LLM calls	35.0k tokens, 6.1 LLM calls	96.0k tokens, 6.6 LLM calls

Table 3: Answer-path differences. All three paths share frozen libraries, question stream, Kimi-K3, and the official scorer.

A REPRODUCIBILITY AND AUDIT DETAILS

A.1 ARTIFACT LEDGER

The machine-readable ledger is generated by the released ledger and recomputation scripts. It records the source summary, dataset, and per-item JSONL SHA-256 values for every run the paper cites, and is regenerated and compared byte-for-byte with the script output before paper writing. The current ledger covers the MemoryAgentBench tracks (two engine generations $\times$ three answer paths, official scorer at commit 455306d, identical `entity2id` fingerprint), LongMemEval full-ingest pairs and the 500-question run, LoCoMo-1986/1540 and the K3-as-answerer diagnostic, LoCoMo-Plus-401, the X7 channel diagnostic, the source-expansion diagnostic, the repaired budget-frontier replay, MEME, BigCodeBench (Zhuo et al., 2025), and ALFWorld (Shridhar et al., 2021). Per-run manifests record dataset and prompt provenance, model identifiers, tool observations, evidence IDs, latency, exit status, and code/runtime hashes; failed integrity checks and negative results are retained rather than dropped. The ledger-generation and recompute scripts ship with the released supplementary material so the chain is regenerable end to end.

The released supplementary material contains the machine-readable manifests, per-item scores, prompts, and runtime hashes needed to regenerate every table. The paper reports dataset, model, judge, subset, and integrity status alongside each result so that the comparisons remain auditable without exposing machine-specific storage locations.

A.2 VERSION PROVENANCE AND ALIGNMENT A/B STUDY

The accompanying evidence package adds a complete three-arm alignment study on LongMemEval-S: three scopes (42+49+46 documents, 137 documents per arm), full ingest, and the same Kimi-K3 endpoint. Arm A is the baseline without window-batch alignment; B1 enables window-batch alignment; B2 additionally enables the revised convergence implementation (parallel alignment,

RQ	Memory capability	Benchmark and observable prediction
1	Adaptive evidence acquisition	MemoryAgentBench compares direct retrieval, a bounded memory-tool loop, and source-grounded verification on the same 767 questions. Scores should separate most on fact consolidation and multi-mention tasks when the agent can keep reading.
2	Temporal and multi-session access	LongMemEval tests updates and cross-session links. Versioned observations should keep both earlier and later facts addressable, while preference questions remain a boundary for latent-user-state modeling.
3	Relational consolidation	LoCoMo separates single-hop, multi-hop, temporal, and open-domain questions. Concept families and relation paths should matter most when an answer joins episodes or revises an earlier value.
4	Revisable construction	Base/Align+ paired runs measure duplicate families, ingest calls, tokens, and task slices. Progressive alignment predicts lower construction cost and better consolidation without requiring a uniform score increase.
5	Deletion and absence boundary	MEME tests deletion, cascade, and absence. The current append-only store should retain provenance but cannot yet guarantee suppression after deletion.

Table 4: Capability-axis evaluation. Each row tests a distinct prediction of the address–evidence separation; published cross-system numbers are kept protocol-labeled rather than folded into a single leaderboard.

Arm	Alignment revision	Calls/doc	Total tokens/doc	Entity redirects	Duplicate groups
A	Baseline	68.4	145,378	0	8
B1	Window-batch alignment	24.2	108,467	1	12
B2	Window-batch + scope-end convergence	29.4	111,853	245	6

Table 5: Three-arm alignment study on 137 LongMemEval-S documents per arm. B1 provides the large efficiency reduction; B2 spends about 3% more tokens than B1 on scope-end convergence and is the only arm with substantial non-destructive family redirection, reducing same-name duplicate groups. The quality slice is too small for a quality claim. B1 and B2 use the same saved service configuration; B2 differs through the runtime implementation and alignment switch.

deferred non-destructive merges, and a scope-end convergence sweep). The saved B1 and B2 service configurations are byte-identical; their difference is in the runtime code and the B2 switch, so the arm assignment is read from the manifest and code snapshot rather than inferred from filenames. The study is powered for efficiency and consolidation, not quality: each scope contributes only one question per evaluation track.

The separate LME Base/Align+ engine diff (initial/revised implementation) is a post-hoc reconstruction from commits *73d9c5e* to *8fa9651*. The latter also contains four throughput fixes, and therefore cannot by itself establish which fixes were active at each point in the Align+ run. We treat the manifest-embedded pipeline configuration, runtime-policy hash, and skill hash as authoritative; the reconstructed diff is provenance for the uncommitted code, not a claim that every later fix was active from the first revised run.

A.3 MEMORYAGENTBENCH SAMPLING FRAME AND CONTAMINATION NOTE

The sampled subset was drawn with a fixed scope list recorded in the released ledger: 767 questions over 10 scopes and 59 documents against the full 3,671 questions over 29 tasks. All four competencies are covered; the omitted tasks are AR’s MH-QA, TTL’s Recommendation, and 19 tasks whose scopes were not sampled. The same subset and `entity2id` fingerprint are used for all six runs (three paths $\times$ two generations). Scope lengths are heterogeneous and disclosed wherever cross-system cells lead: the two FactConsolidation scopes are the 6K variants (26,157 characters each); the 32K/64K/262K FC variants (136k–1.1M characters), the 421K RULER scope, and ten other scopes (17 dropped in total) were dropped for budget, all listed with context sizes in the released ledger’s subset block; EventQA is sampled at the 64K/128K/full-book scopes rather than only the published full-book average; and the LRU domain carries $n=7$ questions (∞ Bench-Sum $n=1$, DetectiveQA $n=6$), so LRU cells are directional. The MAB source corpora are public; whether the answerer’s pretraining saw them cannot be excluded or measured post hoc, and every cross-system table row states its own answerer, judge, and subset.

A.4 ADJUSTED OVERALL DERIVATION

The raw Overall (65.1) averages our four domain scores, but two sampled domains omit tasks that score low for everyone. Imputing the omitted tasks at published non-full-context reference values—

Table 6: X7 provenance ablation, offline retrieval replay on the 210-question gate set. Arms 1–4 form a clean channel ablation: only `enabled_channels` differs, so any recall delta is attributable to that channel set. Arms 5 (evidence-bound agent, gate ON) and 6 (unsurfaced evidence, gate relaxed) reuse arm 4’s five-channel ranking and are retrieval-identical to arm 4; their distinguishing columns (QA accuracy, unsupported-answer rate, mean evidence tokens) require the LLM agent run and are pending. The evidence-gate plumbing was proved offline: with the gate ON, submitting an unsurfaced turn is rejected; with the gate relaxed the same submission is accepted (`plumbing_ok=true`). In the deployed MAB runs the gate fired as designed: 11/767 questions in the deployed Align+ Memory-tool agent run had an evidence submission rejected as unsurfaced (9/767 in Base); 9 of the 11 resubmitted tool-surfaced evidence inside the loop and the remaining 2 ended at the step cap and were answered from tool-surfaced context; zero rejections on the LME runs. Bootstrap CIs: 2,000 resamples, 95%.

Arm	Channels (additive)	Recall-any@10 [95% CI]	Recall-all@10 [95% CI]	Mean bytes@10
1	raw-doc + episode-bm25	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1833
2	+ semantic-provenance	72.38 [66.67, 78.57]	51.43 [44.76, 58.10]	1841
3	+ graph-neighbor	71.43 [65.70, 77.62]	50.95 [44.29, 57.62]	1845
4	+ relation-evidence (full 5)	72.38 [66.65, 78.57]	51.90 [45.24, 58.58]	1847

TTL: our MCC 46 with Recommendation ≈ 13 gives a TTL average ≈ 29.5 ; AR: our three tasks (92 / 88.3 / 98) with MH-QA ≈ 60 gives an AR average ≈ 84.6 ; LRU 41.7 and SF 80.0 unchanged—yields an adjusted Overall of ≈ 59.0 , and varying the imputed values across the published non-FC range gives 59–61. We report the adjusted estimate beside the raw one because the raw one is biased upward by composition; both are stated beside the best published full-context row (49.6), which differs in answerer, judge, and subset—the comparison is contextual, and no claim in the paper subtracts across calibers.

A.5 PER-PATH PROMPTS AND CONFIGURATIONS

All paths use Kimi-K3 as answerer, with their prompts archived verbatim in the run manifests and code snapshots listed in Appendix A.1; the released supplementary material reproduces them in full. The Base→Align+ engine configurations differ as follows: max output tokens 16,384→32,768; usable context window 32k→262,144; retries 2→5; timeout 300 s→1,200 s; and absolute-path persistence enabled in Align+. Direct retrieval makes one fused retrieval call and one answer call; the Memory-tool agent uses a bounded read-only tool loop; the Source-grounded agent materializes a scope, reads and verifies source files, and submits only evidence-gated spans.

A.6 CARRIER ANCHORS (BIGCODEBENCH (ZHUO ET AL., 2025), ALFWORLD (SHRIDHAR ET AL., 2021))

BigCodeBench: 0.4859 completion / 0.2838 hard pass@1 (calibrated split). ALFWorld: 0.9786 in-distribution / 0.9851 out-of-distribution success. Both are memory-insensitive agent-carrier regressions: they anchor that the agent substrate executes tasks correctly, and are not compared horizontally with memory benchmarks.

A.7 MAB INGEST MANIFEST GAPS

The Base and Align+ MAB ingest runs share a 45-document paired intersection. Manifests missing from one generation are excluded from per-document efficiency ratios, which are therefore computed on the intersection only (609→329 calls/doc, −46%); token ratios use the same intersection. Question-level results are unaffected: all 767 questions run against frozen libraries in both generations.

A.8 BUDGET FRONTIER

An earlier depth-10-versus-depth-20 replay is retained as an integrity finding: its prefix invariant failed on all 210 questions (`primary_result_invariant_violations=210`), so it is not used for any positive claim. The repaired design runs retrieval once per query at maximum depth and slices a fixed ranking to $k \in \{1, 3, 5, 10, 20\}$; prefix monotonicity then holds by construction (zero violations), and the frontier rises monotonically from $k=1$ to $k=20$ with per- k percentile-bootstrap CIs recomputed from the frozen replay artifacts.

Benchmark	Protocol	Score	Answerer / judge
LongMemEval-S (500)	sampled, $n=500$	84.60	Kimi-K3; Qwen3.7-plus cross-judge
LongMemEval, direct	paired, $n=25$	0.68 $\rightarrow$ 0.72	Kimi-K3 answerer
LongMemEval, memory-tool	paired, $n=25$	0.68 $\rightarrow$ 0.72	Kimi-K3 answerer
LongMemEval, source-grounded	paired, $n=27$	0.889 $\rightarrow$ 0.926	Kimi-K3; self-judged
LoCoMo-1540	Mem0 binary- J	95.19	Kimi-K3; answerer+judge; $n=1$
LoCoMo-1540 (same answers)	cross-judged	94.22	Qwen3.7-plus; agreement 98.51%

Table 7: Protocol-labeled context on other benchmarks; each row states its answerer and judge. The LoCoMo run differs from Mem0’s published J in two admitted ways—single run versus 10-run mean $\pm$ std, and the answerer/judge model—and carries a disclosed legacy-fingerprint caveat; it is context, not a superiority claim. The LME source-grounded agent track’s two extra questions follow its session-evidence eligibility rule; cross-track comparisons are same-track only.

A.9 X7 CHANNEL ABLATION DETAIL

A.10 PROTOCOL BOUNDARIES

The original LoCoMo token-F1 track includes 1,986 questions and five categories. The Mem0-compatible track contains 1,540 questions and four categories. LongMemEval-S contains 500 questions; the full-ingest paired tracks use $n=25/25/27$ as described in Section 4.1. MEME’s 100 episodes produce heterogeneous task counts (ER/Agg/Tr/Del/Cas/Abs = 100/100/100/100/164/130 after-task checks in the cited run). Raw and real scores are not interchangeable.

A.11 INTEGRITY FAILURES RETAINED

The depth replay of Appendix A.8 reports `primary_result_invariant_violations=210` and `passed=false`. The MEME deletion result is 0% real accuracy. The X7 channel ablation is a null result, reported as an equivalence bound rather than dropped. The ∞ Bench-Sum $n=1$ flip costs the Align+ Overall 2.7 points directly (the net Base $\rightarrow$ Align+ gap is 1.8 points) and is disclosed at every comparison. The LME Direct retrieval–Memory-tool agent tie is reported although it is unflattering to the Memory-tool agent path. These records remain visible because an auditable memory paper should expose failed and null mechanisms as well as successful ones.

A.12 EXTERNAL COMPARISON PROTOCOL

Mem0’s original paper reports LoCoMo category-level F1, BLEU-1, and a binary J in which a GPT-4o-mini judge labels each answer CORRECT or WRONG and the score is the percent judged CORRECT (mean $\pm$ std over 10 independent runs). Zep’s original paper reports DMR and LongMemEval with GPT-4o-family answerers. Rather than re-run those systems under our protocol, we reproduce their published tables verbatim below (Tables 8–10) as cross-protocol contextual references. The Mem0 deterministic token-overlap metrics omitted from the main tables for space are reproduced in Table 12.

The K3-substitution row. The single Deep-Dream J row placed inside Mem0’s J column is produced by running Mem0’s exact binary-judge protocol with Kimi-K3 substituted for GPT-4o-mini as *both* the answerer and the judge—the only change is the model. This yields 95.19% under the same-model K3 judge (97.38/93.62/80.21/95.33 S/M/O/T) and 94.22% under an independent Qwen3.7-plus cross-judge on the same K3 answers (96.91/92.91/77.08/93.46; cross-judge agreement 98.51%). Two differences from Mem0’s published J are admitted: it is a single run rather than a 10-run mean $\pm$ std, and the model is Kimi-K3 rather than GPT-4o-mini. We additionally disclose an integrity caveat: the run manifest carries `diagnostic_override={kind:legacy-missing-runtime-code-fingerprint, formal_claim_allowed:false}`, because 1,529 of the 1,540 answer rows originate from a legacy source run (legacy-source-run) that predates runtime-code fingerprinting. The answers themselves are unchanged (`answer_rows_unchanged=true`, answer SHA-256 76af5772) and the per-item JSONL carries `runtime_code_sha256` on all 2,063 rows (3 distinct hashes: 2fa32774 on 2,019, 36b45933 on 33, 42afb1ac on 11), the same dataset SHA-256 (79fa87e9) and library snapshot (cf2e76b8) as the Qwen-answerer run, and the same Mem0 binary-judge prompt (repository mem0ai/memory-benchmarks, commit 4b61c5d, mode unified-no-evidence-exact-direct). We therefore report 95.19% as a diagnostic cross-judged result, not a fingerprint-complete formal entry.

A stricter controlled comparison—freezing the same answer model, judge model, prompt, temperature, evidence budget, dataset revision, and Graphiti/Zep implementation commit—remains the only path to a shared leaderboard, but it is not a submission hard gate now that the published numbers are cited directly.

A.13 AN INFORMATION-THEORETIC READING OF THE ARCHITECTURE

This appendix gives the paper’s two design commitments a formal shape. Every statement is either a structural property of the architecture (checkable against the schema and code) or a conditional proposition with explicit assumptions; none upgrades the causal wording discipline of Sections 4.2 and 4.8.

A.13.1 WRITE TIME: PROGRESSIVE ENTITY ALIGNMENT

Let the immutable source stream be $S = D_1, \dots, D_T$. Each extracted mention produces a feature vector X_t containing name and content embeddings, attributes, and relation-endpoint co-occurrences. Let Z be the latent entity assignment and let $M_t = (F_t, O_t, G_t)$ be the state after t observations: concept families F_t , an append-only observation store O_t , and a relation graph G_t . An observation is never replaced by a summary; it remains linked to its exact source span.

$$O_{t+1} = O_t \cup \{(X_{t+1}, \text{span}_{t+1})\},$$

$$I(Z; O_{1:t+1}) = I(Z; O_{1:t}) + I(Z; X_{t+1} \mid O_{1:t}) \geq I(Z; O_{1:t}). \quad (7)$$

The second identity is a bookkeeping fact for an append-only stream: new observations can add information about identity, while no prior evidence is discarded. It explains why the system need not solve entity linking on the first mention. A low-confidence merge becomes an additional family (over-splitting); later observations can merge families by redirecting pointers while preserving both observation sets. A premature wrong merge is the dangerous error because it contaminates one address with two entities’ evidence.

Proposition A.1 (Revisability). *In the append-only data model, any coarsening of the current family partition is realizable at a later time by non-destructive merges, with no loss of evidence. If a store instead overwrites an entity with a running summary σ , then $I(Z; \sigma) \leq I(Z; O)$ by the data processing inequality, with strict inequality whenever identity-discriminative detail is discarded.*

Proposition A.2 (Convergence under pooled evidence). *Assume features are i.i.d. conditional on entity identity and two distinct entities $e \neq e'$ emit distributions $P_e \neq P_{e'}$ with Chernoff information $C(P_e, P_{e'}) > 0$. For a same/different-entity test using k pooled observations, the likelihood-ratio error obeys $\Pr[\text{error}] \leq \exp[-kC(P_e, P_{e'})]$ (Chernoff, 1952). Hence the probability that a true merge, initially split under weak evidence, remains undetected falls exponentially with corroborating evidence.*

The assumptions are idealized: real windows are correlated, extraction noise is heavy-tailed, and same-name different-entity pairs can have small C . The prediction is therefore qualitative rather than a measured convergence rate. It matches the implementation’s window-batch adjudication and scope-end convergence pass, which re-test candidate families with more pooled context. In the paired engine runs, same-name duplicate families fall 340→124 (MAB) and 482→99 (LME), while ingest calls per document fall 46–65%; these are the measured anchors, not estimates of C . Relation edges serve two roles: endpoint co-occurrence helps distinguish hypotheses at write time, and the resulting graph supplies additional paths at read time.

A.13.2 READ TIME: HIERARCHICAL ADDRESSES AND PROGRESSIVE DISCLOSURE

The derived layer is an address structure, not an answer reconstruction. Let $A_t(q)$ map a question to concept families and relation neighborhoods, and let ρ be an agent policy that chooses which candidates to open. The resulting source contexts form an adaptive chain

$$C_0(q) \subseteq C_1(q) \subseteq \dots \subseteq S, \quad a_{k+1} = \rho(q, M_t, C_{\leq k}, u_k), \quad (8)$$

where u_k is the agent’s current uncertainty. Concept search gives a fast macro address; family observations and relation edges refine it; the final read returns the exact episode and span. This is why the same representation supports both autonomous exploration and source-grounded answers.

Define reconstruction distortion as the discrepancy between cited text and the source text it claims to represent, and addressing loss as failure to place the relevant source span in C_k . Since every path resolves candidates to verbatim source spans and the gate accepts only spans surfaced in the

current episode, reconstruction distortion is zero by construction; all residual loss is addressing loss. A summarizing memory fixes a rate–distortion point at write time, whereas ρ chooses the reading budget per query (Shannon, 1959). The agent can stop after a compact concept lookup, or disclose more neighborhoods and source files when uncertainty remains, without changing the authority of the evidence.

A.13.3 TWO-RATE DECOMPOSITION AND ANSWER RISK

Let $A(q)$ be the address emitted by the derived overlay and let E_C be the exact source spans selected by a candidate set C . For the answer-relevant variable Y , the chain rule gives

$$I(Y; A, E_C \mid Q = q) = I(Y; A \mid Q = q) + I(Y; E_C \mid A, Q = q). \quad (9)$$

The first term is the information supplied by the compact concept/relation address; the second is the information obtained by opening source text. A summary-only memory replaces S by a channel output $\tilde{S} = f(S)$ and cannot increase $I(Y; \tilde{S} \mid Q)$ beyond $I(Y; S \mid Q)$, whereas Deep-Dream leaves the source channel available after addressing. This is an information-bottleneck interpretation rather than a claim that the implementation solves the bottleneck variational problem (Tishby et al., 2000).

For completeness, define R_q as the event that a relevant source span is addressed and V_q as the event that the verifier answers correctly after reading it. Then

$$\Pr[\hat{Y} = Y \mid q] \geq \Pr[R_q \mid q] \Pr[V_q \mid R_q, q]. \quad (10)$$

The overlay primarily affects address recall, while progressive source disclosure affects conditional verification. The evidence gate sets the probability of a provenance violation to zero under its explicit reachability contract; semantic irrelevance remains outside the guarantee. This factorization explains why improvements should be largest on questions requiring several mentions, temporal updates, or fact consolidation, and why a graph-only offline recall change need not predict the full online gain.

The measured operating points are (25.1k tokens, 0.324), (35.0k, 0.407), and (96.0k, 0.651) Overall on Align+, replicated by Base (0.3025/0.3759/0.6695). The FC-MH scores (2–4 versus 67–70) show where adaptive source reading matters most. X7 bounds the derived channels’ offline retrieval contribution at ± 2.4 points on the development slice; the overlay’s online value is scope selection and link expansion, exercised by the source-grounded agent rather than removed by the ablation.

Proposition A.3 (One-sided addressing error). *For any overlay state, an error in A_t can change which spans enter C_k but cannot change the text or provenance of a span admitted by the evidence gate. Thus overlay errors cause missed coverage, not fabricated cited content; the guarantee is provenance reachability, not semantic entailment.*

This framing adds mathematical structure without upgrading the causal wording of the ladder: the exponential statement is conditional, duplicate-name counts are a proxy, and the measured ladder remains a systems comparison.

Table 8: LoCoMo per-category J (binary CORRECT/WRONG accuracy, LLM-judged). Published rows reproduced verbatim from Chhikara et al. (2025) (arXiv:2504.19413, Table 1): Mem0 reports mean $\pm$ std over 10 runs with GPT-4o-mini as both answerer and judge (std omitted, all ≤ 0.75 ; appendix). **Deep-Dream** applies the *same* J methodology with Kimi-K3 substituted for GPT-4o-mini as both answerer and judge (single run): the primary row uses the same-model judge as Mem0; the $\dagger$ row judges the same K3 answers with an independent Qwen3.7-plus cross-judge (cross-judge agreement 98.51%). Same 841/282/96/321 question populations (1,540 total, Mem0-compatible subset). *Caveat*: the K3-answerer run is flagged `formal_claim_allowed=false`—1,529 historical answer rows predate runtime-code fingerprinting (answers unchanged, sha 76af5772; per-item JSONL carries `runtime_code_sha256`); reported as a diagnostic cross-judge result, not a fingerprint-complete formal result. Deterministic F1/BLEU-1 omitted for space (Table 12). Weighted J : Mem0 ≈ 62.14 , $Mem0^g \approx 61.36$. Deep-Dream agg. J : 95.19% / 94.22% ($\dagger$).

Method	J (S/M/O/T)
LoCoMo	— / — / — / —
ReadAgent	— / — / — / —
MemoryBank	— / — / — / —
MemGPT	— / — / — / —
A-Mem	— / — / — / —
A-Mem*	39.79 / 18.85 / 54.05 / 49.91
LangMem	62.23 / 47.92 / 71.12 / 23.43
Zep	61.70 / 41.35 / 76.60 / 49.31
OpenAI	63.79 / 42.92 / 62.29 / 21.71
Mem0	67.13 / 51.15 / 72.93 / 55.51
$Mem0^g$	65.71 / 47.19 / 75.71 / 58.13
Deep-Dream	97.38 / 93.62 / 80.21 / 95.33
Deep-Dream†	96.91 / 92.91 / 77.08 / 93.46

Table 9: Published Deep Memory Retrieval (DMR) baselines reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). DMR has 500 multi-session conversations (5 sessions, ≤ 12 messages each) with one single-hop fact-retrieval question per conversation. Deep-Dream does not currently run DMR, so this table is a contextual reference only.

Method	Answerer	Accuracy
Recursive Summarization	GPT-4-turbo	35.3%
Conversation Summaries	GPT-4-turbo	78.6%
MemGPT	GPT-4-turbo	93.4%
Full-conversation	GPT-4-turbo	94.4%
Zep	GPT-4-turbo	94.8%
Conversation Summaries	GPT-4o-mini	88.0%
Full-conversation	GPT-4o-mini	98.0%
Zep	GPT-4o-mini	98.2%

Table 10: LongMemEval main results. Published rows (GPT-4o-mini / GPT-4o, Full-context vs. Zep) reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956). The **Deep-Dream** row uses Qwen3.7-plus as both answerer and cross-judge—a different protocol; its accuracy (84.6%) is placed beside the published numbers, not as a shared leaderboard. Latency/avg-context for Deep-Dream are not measured under the published protocol (—). The six-category breakdown is in Table 11.

Answerer	System	Accuracy	Latency (s)	Avg. context
GPT-4o-mini	Full-context	55.4%	31.30	115k
GPT-4o-mini	Zep	63.8%	3.20	1.6k
GPT-4o	Full-context	60.2%	28.90	115k
GPT-4o	Zep	71.2%	2.58	1.6k
Deep-Dream	Qwen3.7-plus	84.6%	—	—

Table 11: LongMemEval six-category breakdown (binary accuracy, %). Published Full-context and Zep values reproduced verbatim from Rasmussen et al. (2025) (arXiv:2501.13956); the **Deep-Dream (Q.)** row is Qwen3.7-plus answerer+cross-judge on the same six categories. Different answerer/judge, so this is a matched-granularity contextual reference, not a shared leaderboard. Sess. pref. is the hardest category for all systems.

System	Sess. pref.	Sess. asst.	Temporal	Multi-sess.	Knowledge upd.	Sess. user
Full-ctx (4o-mini)	30.0	81.8	36.5	40.6	76.9	81.4
Zep (4o-mini)	53.3	75.0	54.1	47.4	74.4	92.9
Full-ctx (4o)	20.0	94.6	45.1	44.3	78.2	81.4
Zep (4o)	56.7	80.4	62.4	57.9	83.3	92.9
Deep-Dream (Q.)	53.3	100.0	84.2	74.4	91.0	98.6

Table 12: Mem0 LoCoMo deterministic metrics, reproduced verbatim from Chhikara et al. (2025) (Table 1). F1 / BLEU-1 over the four categories (Single-hop / Multi-hop / Open-domain / Temporal). Deep-Dream does not compute token-overlap metrics (its metric is binary LLM-judge accuracy, Table 8), so no Deep-Dream column appears here; this table is included for completeness of the published reproduction only.

Method	F1 (S/M/O/T)	BLEU-1 (S/M/O/T)
LoCoMo	25.02 / 12.04 / 40.36 / 18.41	19.75 / 11.16 / 29.05 / 14.77
ReadAgent	9.15 / 5.31 / 9.67 / 12.60	6.48 / 5.12 / 7.66 / 8.87
MemoryBank	5.00 / 5.56 / 6.61 / 9.68	4.77 / 5.94 / 5.16 / 6.99
MemGPT	26.65 / 9.15 / 41.04 / 25.52	17.72 / 7.44 / 34.34 / 19.44
A-Mem	27.02 / 12.14 / 44.65 / 45.85	20.09 / 12.00 / 37.06 / 36.67
A-Mem*	20.76 / 9.22 / 33.34 / 35.40	14.90 / 8.81 / 27.58 / 31.08
LangMem	35.51 / 26.04 / 40.91 / 30.75	26.86 / 22.32 / 33.63 / 25.84
Zep	35.74 / 19.37 / 49.56 / 42.00	23.30 / 14.82 / 38.92 / 34.53
OpenAI	34.30 / 20.09 / 39.31 / 14.04	23.72 / 15.42 / 31.16 / 11.25
Mem0	38.72 / 28.64 / 47.65 / 48.93	27.13 / 21.58 / 38.72 / 40.51
Mem0 ^g	38.09 / 24.32 / 49.27 / 51.55	26.03 / 18.82 / 40.30 / 40.28